DataLab Data Dictionary

Variable Reference Guide

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Overview

This page provides a reference guide for all variables collected in the DataLab survey. Use this when analysing your data or preparing your research reports.

> **Quick Reference**
>
> The data dictionary is organised by station. Each variable includes its type, description, and expected range.

Participant Identifiers

Demographics

Variable	Type	Description	Values
`participant_id`	Integer	Unique participant number	1-60
`lab_room`	Categorical	Lab assignment	304A, 304B
`group_colour`	Categorical	Rotation group	Red, Blue, Green, Yellow, Orange, Purple, Pink, White, Solo
`session_date`	Date	Auto-captured timestamp	ISO format

Sleep & Lifestyle

Baseline

Variable	Type	Description	Range
`sleep_hours`	Continuous	Hours of sleep last night	0-14
`sleep_quality`	Ordinal	Sleep quality rating	1 (Very poor) - 5 (Excellent)
`caffeine_today`	Count	Cups of tea/coffee today	0-10
`exercise_week`	Count	Days exercised this week	0-7
`screen_estimate`	Continuous	Daily phone screen time (hrs)	0-16

Mood (Pre-Post Design)

Pre-Session Baseline

Variable	Type	Description	Range
`mood_pre`	Ordinal	Baseline mood	1-10
`energy_pre`	Ordinal	Baseline energy	1-10
`stress_pre`	Ordinal	Baseline stress	1-10

Post-Session After Stations

Variable	Type	Description	Range
`mood_post`	Ordinal	Post-session mood	1-10
`energy_post`	Ordinal	Post-session energy	1-10
`stress_post`	Ordinal	Post-session stress	1-10

Change Scores Calculated

Variable	Formula	Interpretation
`mood_change`	mood_post - mood_pre	1. **improved**
`energy_change`	energy_post - energy_pre	1. **more energetic**
`stress_change`	stress_post - stress_pre	• **less stressed**

Personality (TIPI)

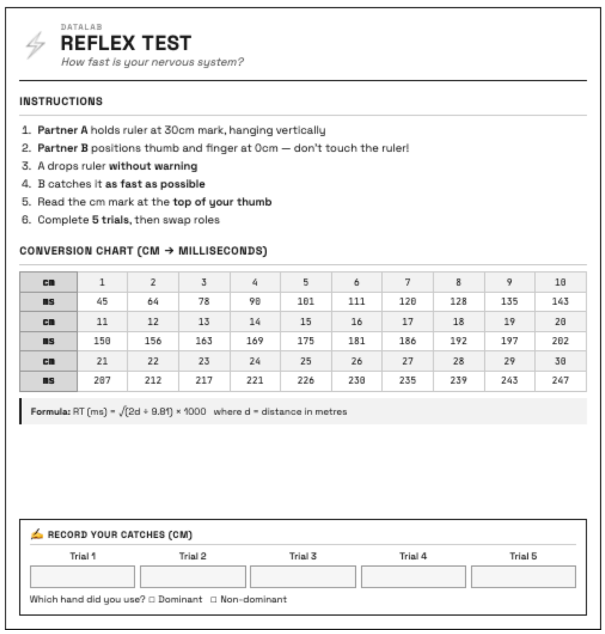
Big Five

The Ten-Item Personality Inventory measures the Big Five personality traits.

Variable	Type	Description	Range
`extraversion`	Continuous	Extraversion score	1-7
`agreeableness`	Continuous	Agreeableness score	1-7
`conscientiousness`	Continuous	Conscientiousness score	1-7
`emotional_stability`	Continuous	Emotional stability score	1-7
`openness`	Continuous	Openness to experience	1-7

Station 1: Reflex Test

Reaction Time



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REFLEX TEST

How fast is your nervous system?

INSTRUCTIONS

1. **Partner A** holds ruler at 30cm mark, hanging vertically
2. **Partner B** positions thumb and finger at 0cm — don't touch the ruler!
3. A drops ruler **without warning**
4. B catches it **as fast as possible**
5. Read the cm mark at the **top of your thumb**
6. Complete **5 trials**, then swap roles

CONVERSION CHART (CM → MILLISECONDS)

cm	1	2	3	4	5	6	7	8	9	10
ms	45	64	78	90	101	111	120	128	135	143
cm	11	12	13	14	15	16	17	18	19	20
ms	150	156	163	169	175	181	186	192	197	202
cm	21	22	23	24	25	26	27	28	29	30
ms	207	212	217	221	226	230	235	239	243	247

Formula: RT (ms) = √(2d ÷ 9.81) × 1000 where d = distance in metres

✍ **RECORD YOUR CATCHES (CM)**

Trial 1	Trial 2	Trial 3	Trial 4	Trial 5

Which hand did you use? ☐ Dominant ☐ Non-dominant

Variable	Type	Description	Range
`rt_hand`	Categorical	Hand used	Dominant, Non-dominant
`rt_trial1_cm` - `rt_trial5_cm`	Continuous	Catch distance per trial	0-30 cm
`rt_missed`	Multi-select	Which trials missed	Trial 1-5, No misses

Calculated Variables

Variable	Formula	Description
`rt_trial1_ms` - `rt_trial5_ms`	$\sqrt{(2d/g)} \times 1000$	RT in milliseconds
`rt_mean_ms`	Average of trials	Mean reaction time
`rt_best_ms`	Minimum of trials	Fastest reaction
`rt_improvement`	Trial 1 - Trial 5	Practice effect

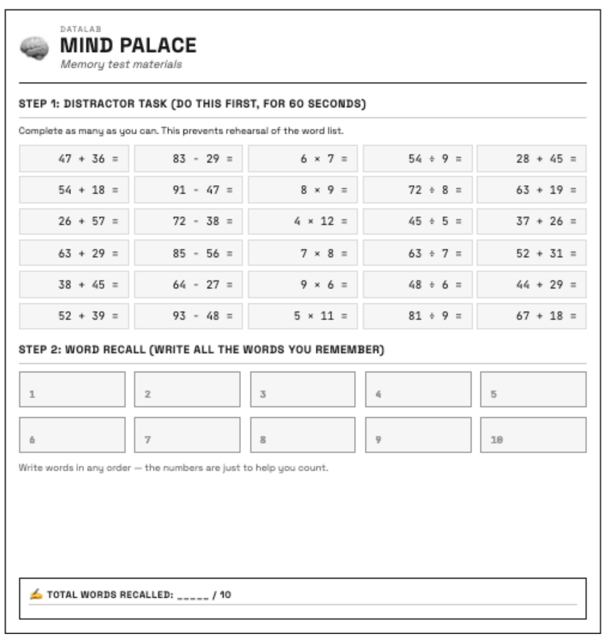
ℹ The Physics

Reaction time (ms) = $\sqrt{(2 \times \text{distance\_cm}/100 \div 9.81)} \times 1000$

A catch at 15cm ≈ 175ms reaction time

Station 2: Mind Palace

Memory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MIND PALACE

Memory test materials

STEP 1: DISTRACTOR TASK (DO THIS FIRST, FOR 60 SECONDS)

Complete as many as you can. This prevents rehearsal of the word list.

47 + 36 =	83 - 29 =	6 × 7 =	54 ÷ 9 =	28 + 45 =
54 + 18 =	91 - 47 =	8 × 9 =	72 ÷ 8 =	63 + 19 =
26 + 57 =	72 - 38 =	4 × 12 =	45 ÷ 5 =	37 + 26 =
63 + 29 =	85 - 56 =	7 × 8 =	63 ÷ 7 =	52 + 31 =
38 + 45 =	64 - 27 =	9 × 6 =	48 ÷ 6 =	44 + 29 =
52 + 39 =	93 - 48 =	5 × 11 =	81 ÷ 9 =	67 + 18 =

STEP 2: WORD RECALL (WRITE ALL THE WORDS YOU REMEMBER)

1	2	3	4	5
6	7	8	9	10

Write words in any order — the numbers are just to help you count.

✍ TOTAL WORDS RECALLED: _____ / 10

Variable	Type	Description	Range
`mem_total`	Count	Words correctly recalled	0-10
`mem_pos1` - `mem_pos10`	Categorical	Recalled each position?	Yes, No, Unsure

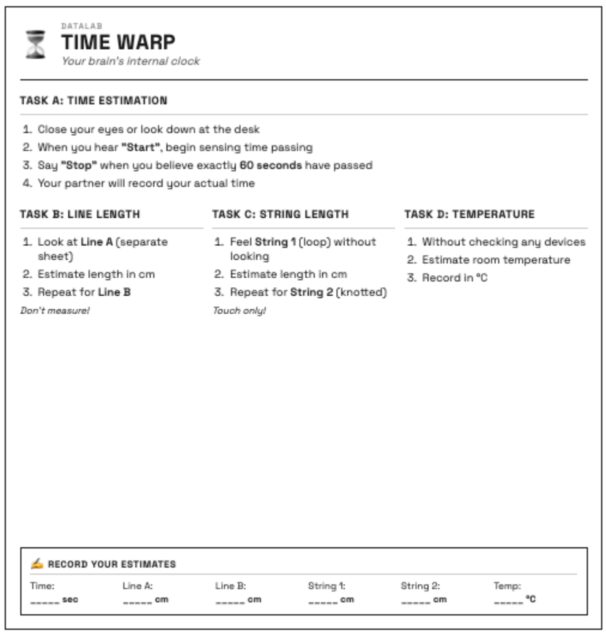
Variable	Type	Description	Range
mem_intrusions	Count	False memories (wrong words)	0-10
mem_strategy	Text	Strategy used	Open text

💡 Serial Position Effect

Positions 1-3 (primacy) and 8-10 (recency) are typically recalled better than positions 4-7 (middle).

Station 3: Time Warp

Estimation



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TIME WARP

Your brain's internal clock

TASK A: TIME ESTIMATION

1. Close your eyes or look down at the desk
2. When you hear **"Start"**, begin sensing time passing
3. Say **"Stop"** when you believe exactly **60 seconds** have passed
4. Your partner will record your actual time

TASK B: LINE LENGTH

1. Look at **Line A** (separate sheet)
2. Estimate length in cm
3. Repeat for **Line B**

Don't measure!

TASK C: STRING LENGTH

1. Feel **String 1** (loop) without looking
2. Estimate length in cm
3. Repeat for **String 2** (knotted)

Touch only!

TASK D: TEMPERATURE

1. Without checking any devices
2. Estimate room temperature
3. Record in °C

✍ RECORD YOUR ESTIMATES

Time: _____ sec | Line A: _____ cm | Line B: _____ cm | String 1: _____ cm | String 2: _____ cm | Temp: _____ °C

Variable	Type	Description	Range
time_estimate_sec	Continuous	60-second estimation	~30-90 sec
line_a_estimate_cm	Continuous	Line A length estimate	1-25 cm
line_b_estimate_cm	Continuous	Line B length estimate	1-25 cm
string1_estimate_cm	Continuous	String 1 estimate (tactile)	1-40 cm
string2_estimate_cm	Continuous	String 2 estimate (tactile)	1-40 cm
temp_estimate_c	Continuous	Room temperature guess	15-30 °C

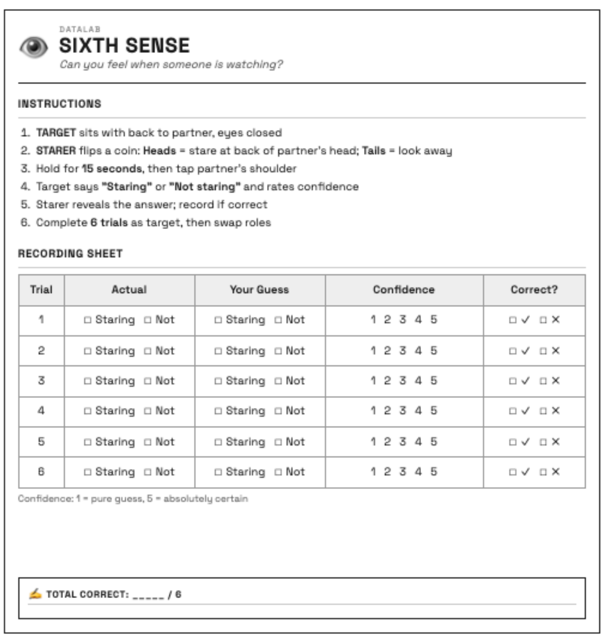
Variable	Type	Description	Range
`time_counting`	Categorical	Counted during timing?	Yes steadily, Yes sometimes, No
`music_condition`	Categorical	Lab condition	silence, music

Calculated Variables

Variable	Formula	Description
`time_error`	estimate - 60	1. **overestimate**
`line_a_error`	estimate - actual	Estimation accuracy
`line_b_error`	estimate - actual	Estimation accuracy
`string1_error`	estimate - actual	Tactile accuracy
`string2_error`	estimate - actual	Tactile accuracy
`temp_error`	estimate - actual	Temperature accuracy

Station 4: Sixth Sense

Detection



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SIXTH SENSE

Can you feel when someone is watching?

INSTRUCTIONS

1. **TARGET** sits with back to partner, eyes closed
2. **STARER** flips a coin: **Heads** = stare at back of partner's head; **Tails** = look away
3. Hold for **15 seconds**, then tap partner's shoulder
4. Target says **"Staring"** or **"Not staring"** and rates confidence
5. Starer reveals the answer; record if correct
6. Complete **6 trials** as target, then swap roles

RECORDING SHEET

Trial	Actual	Your Guess	Confidence	Correct?
1	☐ Staring ☐ Not	☐ Staring ☐ Not	1 2 3 4 5	☐ ✓ ☐ ✕
2	☐ Staring ☐ Not	☐ Staring ☐ Not	1 2 3 4 5	☐ ✓ ☐ ✕
3	☐ Staring ☐ Not	☐ Staring ☐ Not	1 2 3 4 5	☐ ✓ ☐ ✕
4	☐ Staring ☐ Not	☐ Staring ☐ Not	1 2 3 4 5	☐ ✓ ☐ ✕
5	☐ Staring ☐ Not	☐ Staring ☐ Not	1 2 3 4 5	☐ ✓ ☐ ✕
6	☐ Staring ☐ Not	☐ Staring ☐ Not	1 2 3 4 5	☐ ✓ ☐ ✕

Confidence: 1 = pure guess, 5 = absolutely certain

✍ TOTAL CORRECT: _____ / 6

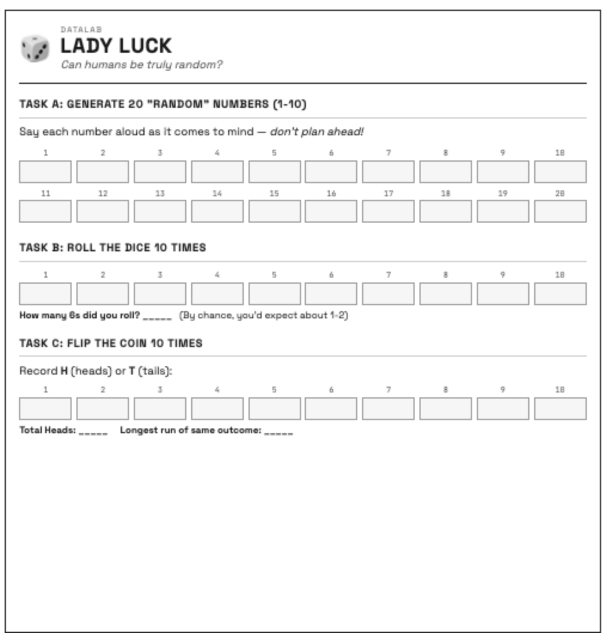
Variable	Type	Description	Values
stare_t1_actual - stare_t6_actual	Categorical	Was partner staring?	Staring, Not staring
stare_t1_guess - stare_t6_guess	Categorical	What you guessed	Staring, Not staring
stare_t1_conf - stare_t6_conf	Ordinal	Confidence	1 (Guessing) - 5 (Certain)
stare_solo	Categorical	Completed solo?	Yes, No

Calculated Variables

Variable	Description	Expected
stare_correct_total	Correct detections (0-6)	~3 (chance)
stare_hit_rate	P(say staring | actually staring)	~0.5
stare_fa_rate	P(say staring | not staring)	~0.5

Station 5: Lady Luck

Randomness



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LADY LUCK

Can humans be truly random?

TASK A: GENERATE 20 "RANDOM" NUMBERS (1-10)

Say each number aloud as it comes to mind — *don't plan ahead!*

1	2	3	4	5	6	7	8	9	10

11	12	13	14	15	16	17	18	19	20

TASK B: ROLL THE DICE 10 TIMES

1	2	3	4	5	6	7	8	9	10

How many 6s did you roll? _____ (By chance, you'd expect about 1-2)

TASK C: FLIP THE COIN 10 TIMES

Record **H** (heads) or **T** (tails):

1	2	3	4	5	6	7	8	9	10

Total Heads: _____ **Longest run of same outcome:** _____

Random Number Generation

Variable	Type	Description	Range
rand_num1 - rand_num20	Integer	Generated "random" numbers	1-10

Dice Rolling

Variable	Type	Description	Range
dice_predict	Categorical	Prediction about 6s	Yes, No, Unsure
dice_roll1 - dice_roll10	Integer	Dice outcomes	1-6

Coin Flipping

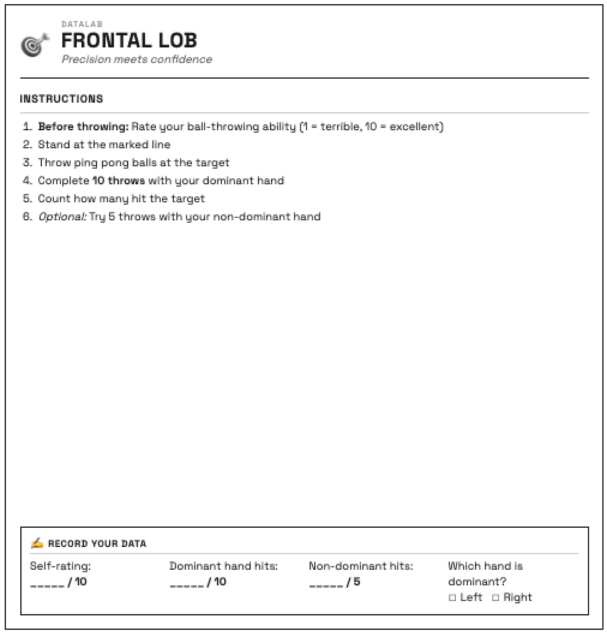
Variable	Type	Description	Range
coin_flip1 - coin_flip10	Categorical	Coin outcomes	Heads, Tails
coin_longest_run	Count	Longest same-outcome streak	1-10

Calculated Variables

Variable	Description	Expected
rand_mean	Mean of 20 numbers	5.5 (true random)
rand_repeats	Adjacent repetitions	~2 (true random)
dice_sixes	Count of 6s rolled	~1.67
coin_heads	Count of heads	~5

Station 6: Frontal Lob

Metacognition



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FRONTAL LOB
Precision meets confidence

INSTRUCTIONS

1. **Before throwing:** Rate your ball-throwing ability (1 = terrible, 10 = excellent)
2. Stand at the marked line
3. Throw ping pong balls at the target
4. Complete **10 throws** with your dominant hand
5. Count how many hit the target
6. *Optional:* Try 5 throws with your non-dominant hand

RECORD YOUR DATA

Self-rating:	Dominant hand hits:	Non-dominant hits:	Which hand is dominant?
_____ / 10	_____ / 10	_____ / 5	☐ Left ☐ Right

Variable	Type	Description	Range
`throw_self_rating`	Ordinal	Self-rated ability (before)	1-10
`throw_dom_success`	Count	Dominant hand successes	0-10
`throw_nondom_success`	Count	Non-dominant successes	0-5
`throw_dom_hand`	Categorical	Dominant hand	Left, Right

Calculated Variables

Variable	Formula	Interpretation
`throw_accuracy_pct`	successes ÷ 10 × 100	Performance %
`throw_calibration`	accuracy - rating	1. **underconfident, - = overconfident**

> **! Calibration Score**
>
> - **Positive**: You performed better than you predicted (underconfident)
> - **Negative**: You predicted better than you performed (overconfident)
> - **Zero**: Perfect calibration!

Session Feedback

End of Session

Variable	Type	Description	Range
`enjoyed_overall`	Ordinal	Overall enjoyment	1-10
`fav_station`	Categorical	Favourite station	1-6
`least_fav_station`	Categorical	Least favourite	1-6
`comments`	Text	Open feedback	Free text

Analysis Ideas

Pre-Post Comparisons (Paired t-tests)

- Did mood improve during DataLab?
- Did stress decrease?
- Did energy change?

Correlations

- Extraversion × mood change
- Sleep hours × energy change
- Conscientiousness × calibration

Group Comparisons (Independent t-tests)

- Lab 304A (music) vs Lab 304B (silence) on time estimation
- Visual vs tactile estimation accuracy

Signal Detection

- Calculate d’ (sensitivity) for stare detection
- Test hit rate against chance (0.5)
